A Bounded Parallel Intersection Type System

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Abstract

We introduce a new presentation of the intersection type discipline in which typing judgments derive *vectors* of types rather than single types. The system uses *binary relations* to control the flow of information between coordinates of these vectors. We refer to this presentation as *system* $\boldsymbol{R}$. The maximal length of type vectors assigned to variables serves as a reasonable notion of *dimension* for system $\mathbf{R}$, which allows for a natural stratification into fragments of bounded dimension.

The present system lies strictly between two known bounded-dimensional systems: the multiset-dimensional system, for which inhabitation is EXPSPACE-complete, and the set-dimensional system, for which inhabitation is undecidable. Our main result is that inhabitation in bounded system $\mathbf{R}$ is decidable in 2-EXPTIME, while for each fixed dimension, inhabitation is decidable in EXPTIME. This result is based on a subformula property restricting the inhabitant search space. Unlike in traditional intersection type systems, the proof of the subformula property requires careful treatment of the additional information flow management capabilities.

Finally, we argue that system $\mathbf{R}$ and its stratification is a valid presentation of the intersection type discipline. First, by proving the subject reduction property for system $\mathbf{R}$ in each bounded dimension, and second, by establishing a correspondence with the classical intersection type system of Barendregt, Coppo, and Dezani-Ciancaglini.

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1 Introduction

Intersection type systems [5, 2] are important in the theory of typed λ-calculi and programming languages [1]. Such systems are highly expressive, characterizing deep semantic properties of λ-terms including normalization and solvability properties [1]. Due to their enormous expressive power, the classical type theoretical decision problems, typability and inhabitation, are undecidable for such systems. It is therefore of interest to investigate bounding principles for intersection type systems in order to obtain decidable fragments, which may be useful for applications (for example, type checking or program synthesis) or interesting for understanding the borderline between decidability and undecidability or the complexity of bounded fragments.

In the present work we focus on the inhabitation problem: Given a type environment and a type, does there exist a term having the type in the environment? Type inhabitation can be understood as the search for a program (term) which satisfies a given specification (type) under given assumptions (type environment), and inhabitation with intersection types may serve as a foundation for program synthesis [11, 16, 9]. Inhabitation understood as proof search has also been used as a guiding principle for the development of Kripke-style semantics for intersection types [10].


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Table 1 Inhabitation complexity for variants of intersection type systems.

Flavor	Restriction	Complexity of Inhabitation
Idempotent	—	undecidable [19]
Idempotent	rank ≥ 3	undecidable [19]
Idempotent	rank 2	EXPSPACE-complete [19]
Non-idempotent (linear)	—	decidable [3], conjectured in NP
Set-dimensional	fixed dimension	unknown [7], conjectured undecidable
Set-dimensional	bounded dimension	undecidable [7]
Multiset-dimensional	fixed dimension	PSPACE-complete [7]
Multiset-dimensional	bounded dimension	EXPSPACE-complete [7]
System **R**	fixed dimension	PSPACE-hard, in EXPTIME (Theorem 38)
System **R**	bounded dimension	EXPSPACE-hard, in 2-EXPTIME (Theorem 37)

A notion of *dimensional bound* was introduced by some of the present authors [7] for intersection type derivations. Intuitively, dimension measures the maximal cumulative width (number of components) of intersection types introduced for subterms in a derivation. Dimension can be measured in two ways, depending on whether width is measured over sets of types or multisets of types. The former corresponds to the standard, idempotent treatment of intersection types, the latter corresponds to a non-idempotent treatment. Relative to each measurement system, we considered inhabitation under dimensional bound: given a type and a dimensional bound, does there exist a term having the type in the given dimension? A key result of [7] is that the dimension-bounded inhabitation problem is undecidable in set-theoretic dimension ([7], Theorem 28) but decidable and EXPSPACE-complete in multiset dimension ([7], Theorem 34). For any fixed dimensional parameter, the latter problem is PSPACE-complete. The main previous result concerning decidable fragments of the inhabitation problem for intersection types had been given by Urzyczyn [19], showing that the problem is decidable and EXPSPACE-complete in rank 2, and undecidable in all ranks k for $k \geq 3$ (here referring to Leivant's notion of rank [13]). All rank 2 typings are derivable in linear multiset dimension [7, Proposition 23] (meaning dimension n for an intersection type with n components). Therefore, bounded multiset dimension strictly generalizes the rank 2 borderline: typings in bounded multiset dimension span all ranks.

In this paper we introduce and study a system (denoted system $\mathbf{R}$[1]) with *bounded parallel intersection* providing a strict generalization of the multiset dimensional system [7], extending (and deepening the analysis of) decidable inhabitation under dimensional bound. Intuitively, system **R** occupies a position between the set-theoretic and the multiset dimensional systems of [7], employing a combination of intersection types represented as vectors of types (which are non-idempotent) and as sets of types (which are idempotent). A technical novelty of the system is the use of *relations* specifying type vector transformations that, intuitively, capture information flow between types in a derivation, and which mediate between vectors of types and sets of types. Dimensionality in this system is measured by the maximal length of vectors of sets of types that can be used to type a variable. Our main technical result is that inhabitation in bounded dimension is decidable in 2-EXPTIME, and for each fixed dimension, inhabitation is decidable in EXPTIME. We have corresponding lower bounds of EXPSPACE and PSPACE, respectively, and leave open the problem of closing the gaps.

[1] The name system **R** has been used by de Carvalho [6] for a different system; we refer to his system as system **Rel**, where relations play a major role in its denotational semantics.

Related work. There are several intersection type systems with features that partially overlap with those of system **R**. However, none of these systems combines a syntax-directed Curry-style presentation with concise information flow management.

So-called non-idempotent intersection type systems have been shown to have decidable inhabitation [3]. These systems are fundamentally different from ours, since they have a linear treatment of type assumptions. Linearity turns out to imply polynomial bounds on the depth of inhabitants (in terms of the size of inhabited types), thereby rendering the inhabitation problem much less complex (in NP) for such systems.

The system λ^{CIL} by Wells et al. [23] (and its simplified version [24]) is a system in Church-style with explicit type annotations in λ-binders where different derivations for the same subterm are managed by means of records. A record of types with labels from I is derived here by a record of terms of corresponding labels from I. Term syntax in λ^{CIL} is enriched with record management capabilities, which we avoid in system **R**.

Similarly to λ^{CIL}, the system by Liquori and Ronchi Della Rocca [14] is a system with explicit annotations at λ-binders. It introduces terms in which variables in binders are annotated with indices. These indices serve as pointers to binders in derivations, and the binders in derivations contain types. Again similarly to λ^{CIL}, term syntax is enriched with type derivation information to establish a Church-style calculus.

The proof system for *Intersection Logic* presented by Pimentel, Ronchi Della Rocca, and Roversi [15] features independent derivations in tuples led according to the same proof rules. Effectively, this works like an intersection type system with vectors. However, the system does not have proof terms and relies on explicit exchange and weakening rules. Another version of the proof system is formulated in terms of a sequent calculus [18].

The line of research on Intersection Logic [17, 18, 15] inspired the development of a vector-based sequent calculus, which we refer to as $(\vdash_{\mathrm{IL}})$, along with corresponding Kripke semantics [10] for Intersection Logic. Proof search in $(\vdash_{\mathrm{IL}})$ corresponds to inhabitation in the system of Barendregt, Coppo and Dezani-Ciancaglini [2]. It is natural to compare fragments of $(\vdash_{\mathrm{IL}})$ in which the length of vectors is bounded to the presented dimensionally bounded system **R**. The most important difference is that in $(\vdash_{\mathrm{IL}})$ reorganization of vector coordinates in assumptions is realized via inverse images of surjective functions (see rule (R) [10, Figure 1]). For example, for the function f such that $f(1) = 1 = f(2)$ the inverse image of the vector $(\{a, b\})$ under f is $f^{-1}((\{a, b\})) = (\{a, b\}, \{a, b\})$. As a result, information can only be duplicated or discarded, and there is no granular control over the *relevant* part of assumptions. Establishing such granular control can be realized via relations, generalizing functions, which is the exact aim of system **R** in the present contribution.

Paper organization. The present work is structured as follows:

Section 2: Definition of system **R** with a dimension bound (Definition 9) and key properties such as weakening (Lemma 16), substitution lemma (Lemma 23), and subject reduction (Lemma 24).

Section 3: Correspondence with the traditional intersection type system of Barendregt, Coppo, and Dezani-Ciancaglini (Theorem 30).

Section 4: 2-EXPTIME decidability of inhabitation in bounded dimension (Theorem 37) and EXPTIME decidability of inhabitation in fixed dimension (Theorem 38).

Section 5: Comparison with the multiset-dimensional system (Lemma 39, Lemma 41, and Lemma 42) and EXPSPACE-hardness of inhabitation in bounded dimension (Corollary 40).

Section 6: Concluding remarks.

2 System R

We denote *intersection types* by A, B, finite sets of intersection types by σ, τ, and *type atoms* by a, b.

▶ **Definition 1** (Intersection Types).

$$\begin{array}{rcl} A, B & ::= & a \mid \sigma \to A \\ \sigma, \tau & ::= & \{A_1, \ldots, A_n\} \text{ where } n \geq 0 \end{array}$$

Intersection types are a variant of strict types considered by van Bakel [20, 21, 22].

We denote vectors of types and vectors of sets of types by $\bar{A}$ and $\bar{\sigma}$ respectively. If $\bar{\sigma} = (\sigma_1, \ldots, \sigma_n)$ and $\bar{A} = (A_1, \ldots, A_n)$, then $\bar{\sigma} \Rightarrow \bar{A} = (\sigma_1 \to A_1, \ldots, \sigma_n \to A_n)$, where $n \geq 0$, and in particular $() \Rightarrow () = ()$. We write $\bar{A}_{[i]}$ for A_i and $\bar{\sigma}_{[i]}$ for σ_i. The *dimension* of a vector is the number of its *coordinates*, that is $\dim(A_1, \ldots, A_n) = n$. We extend relations $\subseteq$, $\in$, and the operation $\cup$ to vectors.

▶ **Definition 2** (Vector Extension).

- $(\sigma_1, \ldots, \sigma_n) \mathbin{\bar{\subseteq}} (\tau_1, \ldots, \tau_n)$ *if* $\sigma_1 \subseteq \tau_1, \ldots, \sigma_n \subseteq \tau_n$
- $(A_1, \ldots, A_n) \mathbin{\bar{\in}} (\sigma_1, \ldots, \sigma_n)$ *if* $A_1 \in \sigma_1, \ldots, A_n \in \sigma_n$
- $(\sigma_1, \ldots, \sigma_n) \mathbin{\bar{\cup}} (\tau_1, \ldots, \tau_n) := (\sigma_1 \cup \tau_1, \ldots, \sigma_n \cup \tau_n)$
- $(\{A_1\}, \ldots, \{A_n\})\downarrow := (A_1, \ldots, A_n)$

▶ **Definition 3** (Environment of Dimension d). *An* environment Σ *is a finite set of* type assumptions *having the shape* $x : \sigma$ *for distinct term variables. An* environment Γ *of dimension* d *is a finite set of* type assumptions *having the shape* $x : \bar{\sigma}$ *for distinct term variables such that* $\dim(\bar{\sigma}) = d$.

Each environment Γ is equipped with its dimension $\dim(\Gamma)$. In particular, the empty environment of dimension 0 is distinct from the empty environment of dimension 1.

We may extend an environment Γ by an additional assumption $x : \bar{\sigma}$, written $\Gamma, x : \bar{\sigma}$, where $\dim(\bar{\sigma}) = \dim(\Gamma)$ and x does not appear in any assumption in Γ. The *domain* of an environment Γ is given by $\mathrm{dom}(\Gamma) = \{x \mid (x : \bar{\sigma}) \in \Gamma \text{ and } \bar{\sigma} \neq (\emptyset, \ldots, \emptyset)\}$. When $(x : \bar{\sigma}) \in \Gamma$ we write $\Gamma(x)$ for $\bar{\sigma}$. We identify Γ and $\Gamma, x : (\emptyset, \ldots, \emptyset)$, consequently $\Gamma(y) = (\emptyset, \ldots, \emptyset)$ iff $y \notin \mathrm{dom}(\Gamma)$ for any term variable y.

Similarly to vectors, we extend the relation $\subseteq$, the operation $\cup$, and coordinate indexing to environments.

▶ **Definition 4** (Environment Extension). *Let* Γ, Δ *be environments such that* $\dim(\Gamma) = \dim(\Delta)$.

- $\Gamma \subseteq \Delta$ *if* $\Gamma(x) \mathbin{\bar{\subseteq}} \Delta(x)$ *for* $x \in \mathrm{dom}(\Gamma)$
- $(\Gamma \cup \Delta)(x) := \Gamma(x) \mathbin{\bar{\cup}} \Delta(x)$
- $\Gamma_{[i]} := \{x : \bar{\sigma}_{[i]} \mid (x : \bar{\sigma}) \in \Gamma\}$

The original feature of the system presented in this paper is the management of flow of types between vectors by means of relations. We extend notation of previous work [10] from surjective functions to binary relations, and define type vector transformations as follows.

▶ **Definition 5** (Type Vector Transformation). *Let* $R \subseteq \{1, \ldots, n\} \times \{1, \ldots, m\}$ *be a binary relation where* $n, m \geq 0$.

- $R((A_1, \ldots, A_n)) := (\tau_1, \ldots, \tau_m)$ *where* $\tau_i = \{A_j \mid j \, R \, i\}$
- $R((\sigma_1, \ldots, \sigma_n)) := (\tau_1, \ldots, \tau_m)$ *where* $\tau_i = \bigcup\{\sigma_j \mid j \, R \, i\}$
- $R(\{x_1 : \bar{\sigma}_1, \ldots, x_l : \bar{\sigma}_l\}) := \{x_1 : R(\bar{\sigma}_1), \ldots, x_l : R(\bar{\sigma}_l)\}$

▶ **Example 6.** For $R = \{(1,1), (1,3), (2,1), (3,3)\}$ we have $R((a,b,c)) = (\{a,b\}, \{\}, \{a,c\})$.

▶ **Definition 7.** *Let $R \subseteq \{1,\dots,n\} \times \{1,\dots,m\}$ be a relation.*

- *R is* left-total *when for each $i \in \{1,\dots,n\}$ there is a j such that $i\,R\,j$*
- *R is* right-total *when for each $j \in \{1,\dots,m\}$ there is an i such that $i\,R\,j$*
- *R is* left-unique *when for each $i \in \{1,\dots,n\}$ there is at most one j such that $i\,R\,j$*
- *R is* right-unique *when for each $j \in \{1,\dots,m\}$ there is at most one i such that $i\,R\,j$*

If a relation R is right-unique and right-total, then the vector of types $R(\bar{B})\!\downarrow$ is well-defined. Primary properties of vector transformations are listed in the following Lemma 8.

▶ **Lemma 8** (Properties of Vector Transformations)**.**

1. *For two binary relations $R, R' \subseteq \{1,\dots,n\} \times \{1,\dots,m\}$ it holds that if $R \subseteq R'$, then for all $\bar{\sigma}$ such that $\dim(\bar{\sigma}) = n$ we have $R(\bar{\sigma}) \mathrel{\bar{\subseteq}} R'(\bar{\sigma})$.*
2. *If $R(\bar{\sigma}) \mathrel{\bar{\subseteq}} R'(\bar{A})$ and R is left-total, then there exists R'' such that $R''(\bar{A}) = \bar{\sigma}$.*
3. *If $\bar{A}$ has distinct coordinates, $R''(\bar{A}) = \bar{\sigma}$, $R(\bar{\sigma}) \mathrel{\bar{\subseteq}} R'(\bar{A})$, and R is left-total, then $R \circ R'' \subseteq R'$.*
4. *$R(\Gamma_1 \cup \Gamma_2) = R(\Gamma_1) \cup R(\Gamma_2)$*
5. *If R is right-unique and right-total, and $\bar{A} \mathrel{\bar{\in}} \bar{\sigma}$ then $R(\bar{A})\!\downarrow \mathrel{\bar{\in}} R(\bar{\sigma})$.*
6. *If R is right-unique and right-total, then $R(\bar{\sigma} \Rightarrow \bar{A})\!\downarrow = R(\bar{\sigma}) \Rightarrow R(\bar{A})\!\downarrow$.*
7. *If $R(\bar{\sigma}) \mathrel{\bar{\subseteq}} \bar{\tau}$ and R is left-total, then there exists $\bar{\sigma}'$ such that $\bar{\sigma} \mathrel{\bar{\subseteq}} \bar{\sigma}'$, $R(\bar{\sigma}') \mathrel{\bar{\subseteq}} \bar{\tau}$, and for all $i \in \{1,\dots,\dim(\bar{\sigma}')\}$ there exists $j \in \{1,\dots,\dim(\bar{\tau})\}$ such that $\bar{\sigma}'_{[i]} = \bar{\tau}_{[j]}$.*

Proof.

1. The claim follows from $R(\bar{\sigma})_{[i]} = \bigcup\{\bar{\sigma}_{[j]} \mid j\,R\,i\} \subseteq \bigcup\{\bar{\sigma}_{[j]} \mid j\,R'\,i\} = R'(\bar{\sigma})_{[i]}$.
2. Consider $R'' = \{(i,j) \mid \exists k.i\,R'\,k \text{ and } j\,R\,k \text{ and } \bar{A}_{[i]} \in \bar{\sigma}_{[j]}\}$. We have $R''(\bar{A})_{[j]} = \{\bar{A}_{[i]} \mid i\,R''\,j\} = \{\bar{A}_{[i]} \mid \exists k.i\,R'\,k \text{ and } j\,R\,k \text{ and } \bar{A}_{[i]} \in \bar{\sigma}_{[j]}\}$. This set is included in $\bar{\sigma}_{[j]}$ as for each its element $\bar{A}_{[i]}$ we have $\bar{A}_{[i]} \in \bar{\sigma}_{[j]}$. Similarly, $\bar{\sigma}_{[j]} \subseteq R(\bar{\sigma})_{[k]}$ for some k as R is left-total. As $R(\bar{\sigma})_{[k]} \subseteq R'(\bar{A})_{[k]}$, each element of $\bar{\sigma}_{[j]}$ must be equal to some $\bar{A}_{[i]}$ such that $i\,R'\,k$.
3. We have $(R \circ R'')(\bar{A}) = R(\bar{\sigma}) \mathrel{\bar{\subseteq}} R'(\bar{A})$. Since $\bar{A}$ has distinct coordinates, we obtain the claim.
4. Immediate from the definition.
5. Since R is right-unique and right-total, for each $i \in \{1,\dots,\dim(R(\bar{A})\!\downarrow)\}$ there is exactly one j such that $j\,R\,i$, and we have $(R(\bar{A})\!\downarrow)_{[i]} = \bar{A}_{[j]} \in \bar{\sigma}_{[j]} \subseteq R(\bar{\sigma})_{[i]}$.
6. Since R is right-unique and right-total, we have $(R(\bar{\sigma} \Rightarrow \bar{A})\!\downarrow)_{[i]} = \bar{\sigma}_{[j]} \to \bar{A}_{[j]}$ where $j\,R\,i$ for exactly one j. Similarly, $R(\bar{\sigma})_{[i]} = \bar{\sigma}_{[j]}$ and $(R(\bar{A})\!\downarrow)_{[i]} = \bar{A}_{[j]}$ where $j\,R\,i$. Consequently, we obtain $R(\bar{\sigma} \Rightarrow \bar{A})\!\downarrow = R(\bar{\sigma}) \Rightarrow R(\bar{A})\!\downarrow$.
7. Since R is left-total, for all $i \in \{1,\dots,\dim(\bar{\sigma})\}$ there exists $j \in \{1,\dots,\dim(\bar{\tau})\}$ such that $i\,R\,j$, for which we set $\bar{\sigma}'_{[i]} := \bar{\tau}_{[j]}$, satisfying the required properties. ◀

Judgments in the following *system* $\boldsymbol{R}$ with *dimension bound* k are of shape $\Gamma \vdash_k t : \bar{A}$ such that $\dim(\Gamma) = \dim(\bar{A})$.

▶ **Definition 9** (System **R** with Dimension Bound k)**.**

$$\frac{\bar{A} \mathrel{\bar{\in}} \bar{\sigma} \qquad \dim(\Gamma) = \dim(\bar{A}) \le k}{\Gamma, x : \bar{\sigma} \vdash_k x : \bar{A}}\ (\mathrm{Ax}) \qquad \frac{\Gamma, x : \bar{\sigma} \vdash_k t : \bar{A}}{\Gamma \vdash_k \lambda x.t : \bar{\sigma} \Rightarrow \bar{A}}\ (\Rightarrow\mathrm{I})$$

$$\frac{\Gamma \vdash_k t : R(\bar{A}) \Rightarrow \bar{B} \qquad \Delta \vdash_k u : \bar{A} \qquad R \subseteq \{1,\dots,\dim(\Delta)\} \times \{1,\dots,\dim(\Gamma)\}}{\Gamma \cup R(\Delta) \vdash_k t\ u : \bar{B}}\ (\Rightarrow\mathrm{E})$$

We may leave out the actual bound, if it is immaterial.

▶ **Definition 10** (System **R**)**.** $\Gamma \vdash t : \bar{A}$ *means* $\Gamma \vdash_k t : \bar{A}$ *for some* k.

▶ **Lemma 11.** *The following rule is derivable:* $\dfrac{}{\emptyset \vdash_k t : ()}\ (nil)$.

Proof. We proceed by induction on t.
Case $t = x$: Due to $\emptyset, x : () = \emptyset$, by rule (Ax) we have $\emptyset \vdash_k x : ()$.
Case $t = \lambda x.u$: By the induction hypothesis we have $\emptyset \vdash_k u : ()$. Due to $\emptyset, x : () = \emptyset$ and $() \Rightarrow () = ()$, by rule ($\Rightarrow$I) we have $\emptyset \vdash_k \lambda x.u : ()$.
Case $t = u\,v$: By the induction hypothesis we have $\emptyset \vdash_k u : ()$ and $\emptyset \vdash_k v : ()$. For $R = \emptyset$ we have $() = R(()) \Rightarrow ()$ and $\emptyset \cup R(\emptyset) = \emptyset$. Therefore, by rule ($\Rightarrow$E) we have $\emptyset \vdash_k u\,v : ()$. ◀

Since coordinates in assumptions are sets, system **R** is not linear, illustrated by derivability of $\{x : (\{a\}), y : (\{\{a\} \to \{a\} \to b\})\} \vdash_1 y\,x\,x : (b)$. Complementarily, the following example shows that the occurrence of sets of types with cardinality of more than one in derivations does not necessarily increase dimension.

▶ **Example 12.**
We have $\emptyset \vdash_1 \lambda x.x\,x : (\{a, \{a\} \to b\} \to b)$ by the following derivation, omitting rule (Ax) and using $R := \{(1,1)\}$:

$$\dfrac{\dfrac{\{x : (\{\{a\} \to b\})\} \vdash_1 x : R((a)) \Rightarrow (b) \qquad \{x : (\{a\})\} \vdash_1 x : (a)}{\{x : (\{a, \{a\} \to b\})\} \vdash_1 x\,x : (b)}\ (\Rightarrow\text{E})}{\emptyset \vdash_1 \lambda x.x\,x : (\{a, \{a\} \to b\} \to b)}\ (\Rightarrow\text{I})$$

System **R** does not have an explicit intersection introduction rule. Rather, relations in the arrow elimination rule are the mechanism by which intersection introduction is managed within a syntax-directed, Curry-style presentation. The following example illustrates the action of different relations in one derivation.

▶ **Example 13.**
For the following definitions
- $\Gamma_1 := \{x : (\{\{b, c\} \to a\})\}$
- $\Gamma_2 := \{y : (\{\{a\} \to b\}, \{\{a\} \to c\})\}$
- $\Gamma_3 := \{x : (\{\{b, c\} \to a\}), y : (\{\{a\} \to b, \{a\} \to c\}), z : (\{a\})\}$
- $R_1 := \{(1,1), (2,1)\}$ with $R_1((b,c)) = (\{b, c\})$
- $R_2 := \{(1,1), (1,2)\}$ with $R_2((a)) = (\{a\}, \{a\})$

we have the following derivation:

$$\dfrac{\Gamma_1 \vdash_2 x : R_1((b,c)) \Rightarrow (a) \qquad \Gamma_2 \cup R_2(\{z : (\{a\})\}) \vdash_2 y\,z : (b,c)}{\Gamma_3 \vdash_2 x\,(y\,z) : (a)}\ (\Rightarrow\text{E})$$

for which the judgment $\Gamma_2 \cup R_2(\{z : (\{a\})\}) \vdash_2 y\,z : (b,c)$ is derived in dimension 2 as follows:

$$\dfrac{\Gamma_2 \vdash_2 y : R_2((a)) \Rightarrow (b,c) \qquad \{z : (\{a\})\} \vdash_2 z : (a)}{\Gamma_2 \cup R_2(\{z : (\{a\})\}) \vdash_2 y\,z : (b,c)}\ (\Rightarrow\text{E})$$

In the above rule applications, the role of R_1 is to combine types in coordinates into an intersection, and the role of R_2 is to copy a type to two coordinates.

The dimension bound limits the length of vectors occurring in a derivation, and therefore can be weakened to any larger bound.

▶ **Lemma 14** (Dimensional Restriction)**.** *Any vector occurring in a derivation of* $\Gamma \vdash_k t : \bar{A}$ *has at most* k *coordinates.*

Proof. Induction on t. ◀

▶ **Lemma 15** (Dimension Weakening). *If* $\Gamma \vdash_k t : \bar{A}$ *and* $k \leq m$ *then* $\Gamma \vdash_m t : \bar{A}$.

Proof. Induction on t. For rule (Ax) the dimension restriction $n \leq k$ is replaced by $n \leq m$, which holds given $k \leq m$. ◀

Any type assumption can be weakened (increasing the cardinality of the corresponding assigned set of types) with no effect on dimension.

▶ **Lemma 16** (Weakening). *If* $\Gamma \vdash_k t : \bar{A}$ *and* $\Gamma \subseteq \Delta$, *then* $\Delta \vdash_k t : \bar{A}$.

Proof. Assuming $\Gamma \vdash_k t : \bar{A}$ and $\Gamma \subseteq \Delta$, we proceed by induction on t.
Case $t = x$: Due to $\bar{A} \mathrel{\bar{\in}} \Gamma(x) \mathrel{\bar{\subseteq}} \Delta(x)$ we obtain the claim by rule (Ax).
Case $t = \lambda x.u$: For some $\bar{\sigma}$ and $\bar{B}$ we have $\bar{A} = \bar{\sigma} \Rightarrow \bar{B}$, and by the induction hypothesis we have $\Delta, x : \bar{\sigma} \vdash_k u : \bar{B}$. By rule ($\Rightarrow$I) we obtain the claim.
Case $t = u\,v$: For some Δ', Γ', $\bar{B}$, and R such that $\Gamma = \Gamma' \cup R(\Delta')$ we have $\Gamma' \vdash_k u : R(\bar{B}) \Rightarrow \bar{A}$ and $\Delta' \vdash_k v : \bar{B}$. Due to $\Gamma' \subseteq \Gamma \subseteq \Delta$ by the induction hypothesis we have $\Delta \vdash_k u : R(\bar{B}) \Rightarrow \bar{A}$. By rule ($\Rightarrow$E) we have $\Delta \cup R(\Delta') \vdash_k u\,v : \bar{A}$. Due to $R(\Delta') \subseteq \Gamma \subseteq \Delta$ we have $\Delta \cup R(\Delta') = \Delta$, obtaining the claim. ◀

A key technical insight for system **R** is the following Lemma 17, which allows us to permute, remove, or copy vector coordinates in derivations.

▶ **Lemma 17.** *If* $\Gamma \vdash_k t : \bar{A}$ *and* $R \subseteq \{1, \ldots, \dim(\bar{A})\} \times \{1, \ldots, m\}$ *is right-unique and right-total, then* $R(\Gamma) \vdash_{\max(k,m)} t : R(\bar{A})\!\downarrow$.

Proof. Assuming $\Gamma \vdash_k t : \bar{A}$ and that R is right-unique and right-total, we proceed by induction on t. Let $n = \max(k, m)$.
Case $t = x$: We have $\bar{A} \mathrel{\bar{\in}} \Gamma(x)$. By Lemma 8(5), we have $R(\bar{A})\!\downarrow \mathrel{\bar{\in}} R(\Gamma(x))$, which by rule (Ax) implies $R(\Gamma) \vdash_n x : R(\bar{A})\!\downarrow$.
Case $t = \lambda x.u$: For some $\bar{\sigma}$ and $\bar{B}$ we have $\bar{A} = \bar{\sigma} \Rightarrow \bar{B}$, and by the induction hypothesis we have $R(\Gamma), x : R(\bar{\sigma}) \vdash_n u : R(\bar{B})\!\downarrow$. By Lemma 8(6) we have $R(\bar{\sigma}) \Rightarrow R(\bar{B})\!\downarrow = R(\bar{\sigma} \Rightarrow \bar{B})\!\downarrow = R(\bar{A})\!\downarrow$. By rule ($\Rightarrow$I) we obtain $R(\Gamma) \vdash_n \lambda x.u : R(\bar{A})\!\downarrow$.
Case $t = u\,v$: For some Δ, R', and $\bar{B}$ we have $\Delta \vdash_k v : \bar{B}$, and weakening the environment by Lemma 16, we have $\Gamma \vdash_n u : R'(\bar{B}) \Rightarrow \bar{A}$. By the induction hypothesis we obtain $R(\Gamma) \vdash_n u : R(R'(\bar{B}) \Rightarrow \bar{A})\!\downarrow$. By Lemma 15 we have $\Delta \vdash_n v : \bar{B}$. Let $R'' = R \circ R'$. By Lemma 8(6) we have $R(R'(\bar{B}) \Rightarrow \bar{A})\!\downarrow = R''(\bar{B}) \Rightarrow R(\bar{A})\!\downarrow$. By Lemma 8(4) we have $R(\Gamma) \cup R''(\Delta) = R(\Gamma \cup R'(\Delta))$. By rule ($\Rightarrow$E) we obtain $R(\Gamma \cup R'(\Delta)) \vdash_n u\,v : R(\bar{A})\!\downarrow$. ◀

The following Example 18 illustrates use of the above lemma to permute, remove, or copy vector coordinates in derivations.

▶ **Example 18.** Assume $\Gamma \vdash_k t : (A_1, A_2, A_3)$. By Lemma 17 we have:
- $R_1(\Gamma) \vdash_{\max(k,2)} t : (A_3, A_1)$ where $R_1 := \{(1, 2), (3, 1)\}$
- $R_2(\Gamma) \vdash_{\max(k,4)} t : (A_1, A_2, A_2, A_3)$ where $R_2 := \{(1, 1), (2, 2), (2, 3), (3, 4)\}$

A subtle aspect of Lemma 17 is that coordinate copies do not increase the dimension besides the space required for their occurrence. In particular, occurring relations can be adjusted to accommodate additional copies.

▶ **Example 19.** Let $\sigma := \{a, b, c\}$. We have $\{x : (\sigma), y : (\{\sigma \to d\})\} \vdash_3 y\,x : (d)$, and 3 is the minimal dimension necessary to derive this judgment. By Lemma 17, using the relation $\{(1, 1), (1, 2)\}$ we have $\{x : (\sigma, \sigma), y : (\{\sigma \to d\}, \{\sigma \to d\})\} \vdash_3 y\,x : (d, d)$, essentially copying the derivation. Still, dimension 3 suffices to account for the copied coordinate.

Let us discuss the properties of the dimension bound with respect to different notions of reduction. The dimension bound is neither invariant under β-expansion (Example 20), η-expansion (Example 21), nor under η-reduction (Example 22).

▶ **Example 20.** We have $\emptyset \vdash_1 (\lambda x.x)\,(\lambda x.x) : (\{a\} \to a)$ in bounded dimension 1 and $(\lambda y.y\,y)\,(\lambda x.x) \to_\beta (\lambda x.x)\,(\lambda x.x)$. However, $\emptyset \vdash_k (\lambda y.y\,y)\,(\lambda x.x) : (\{a\} \to a)$ requires $k \geq 2$. This matches the understanding that typing a non-linear β-redex in an intersection type system may require intersection introduction.

▶ **Example 21.** We have $\emptyset \vdash_1 \lambda x.x : (\{\{a,b\} \to c\} \to \{a,b\} \to c)$ in bounded dimension 1 and $\lambda xy.x\,y \to_\eta \lambda x.x$. However, $\emptyset \vdash_k \lambda xy.x\,y : (\{\{a,b\} \to c\} \to \{a,b\} \to c)$ requires $k \geq 2$.

▶ **Example 22.** We have $\emptyset \vdash_1 \lambda xy.x\,y : (\{\{a\} \to c\} \to \{a,b\} \to c)$ and $\lambda xy.x\,y \to_\eta \lambda x.x$. However, $\emptyset \nvdash_k \lambda x.x : (\{\{a\} \to c\} \to \{a,b\} \to c)$ for any dimension bound k.

The above Example 21 and Example 22 are coherent with intersection type systems which do not include subtyping and do require intersection introduction in order to assign an intersection of types (cf. Theorem 28 and the line of work by van Bakel [21, 22]).

Let us conclude this section by showing that for each dimension bound subject reduction with respect to β-reduction holds. This renders each dimensionally bounded fragment a coherent type system. The result is achieved by careful management of information flow in the proof of the following substitution lemma. The crucial aspect of the proof is an appropriate choice of a relation in the application case.

▶ **Lemma 23** (Substitution Lemma). *If* $\Gamma, x : R(\bar{A}) \vdash_k t : \bar{B}$ *and* $\Delta \vdash_k u : \bar{A}$, *then* $\Gamma \cup R(\Delta) \vdash_k t[x := u] : \bar{B}$.

Proof. Assuming $\Gamma, x : R(\bar{A}) \vdash_k t : \bar{B}$ and $\Delta \vdash_k u : \bar{A}$, we proceed by induction on t. We also note here that $\Gamma, x : R(\bar{A}) \vdash_k t : \bar{B}$ implies that $R \subseteq \{1, \dots, \dim(\bar{A})\} \times \{1, \dots, m\}$ where $m \leq k$. W.l.o.g. $\bar{A}$ has distinct coordinates as we can remove duplicate ones by Lemma 17.

Case $t = x$: We have $\bar{B}_{[i]} \in R(\bar{A})_{[i]}$ for $i \in \{1, \dots, \dim(\bar{B})\}$. Therefore, there is a right-unique $R' \subseteq R$ such that $R'(\bar{A}) = \bar{B}$. By Lemma 17 we have $R'(\Delta) \vdash_k u : R'(\bar{A})\!\downarrow$. Due to $R' \subseteq R$, we obtain $\Gamma \cup R(\Delta) \vdash_k u : \bar{B}$ by properties of vector transformations and the weakening lemma (cf. Lemma 8(1) and Lemma 16).

Case $t = y \neq x$: We have $\Gamma \vdash_k y : \bar{B}$ and the claim follows by the weakening lemma (cf. Lemma 16).

Case $t = \lambda y.v$: We may assume due to α-conversion that $y \neq x$ and $y \notin \mathrm{var}(u)$. For some $\bar{\sigma}$ and $\bar{C}$ we have $\bar{B} = \bar{\sigma} \Rightarrow \bar{C}$ and $\Gamma, x : R(\bar{A}), y : \bar{\sigma} \vdash_k v : \bar{C}$. By the induction hypothesis we obtain $\Gamma \cup R(\Delta), y : \bar{\sigma} \vdash_k v[x := u] : \bar{C}$. The claim follows by rule ($\Rightarrow$I).

Case $t = v\,w$: For some Γ', Δ', R', $\bar{\sigma}$, $\bar{\tau}$, and $\bar{C}$ such that $\Gamma = \Gamma' \cup R'(\Delta')$ and $R(\bar{A}) = \bar{\tau} \sqcup R'(\bar{\sigma})$ we have $\Gamma', x : \bar{\tau} \vdash_k v : R'(\bar{C}) \Rightarrow \bar{B}$ and $\Delta', x : \bar{\sigma} \vdash_k w : \bar{C}$.
Note that $R' \subseteq \{1, \dots, \dim(\bar{C})\} \times \{1, \dots, \dim(\bar{B})\}$. W.l.o.g. R' is left-total. In order to obtain the claim by rule ($\Rightarrow$E) we consider the corresponding premises.

- Due to $\bar{\tau} \sqsubseteq R(\bar{A})$, by the weakening lemma (cf. Lemma 16) and the induction hypothesis we have $\Gamma' \cup R(\Delta) \vdash_k v[x := u] : R'(\bar{C}) \Rightarrow \bar{B}$.
- Due to $R'(\bar{\sigma}) \sqsubseteq R(\bar{A})$ and since R' is left-total and $\bar{A}$ has distinct coordinates, by Lemma 8(2) and Lemma 8(3) there exists a R'' such that $R' \circ R'' \subseteq R$ and $R''(\bar{A}) = \bar{\sigma}$. By the induction hypothesis we have $\Delta' \cup R''(\Delta) \vdash_k w[x := u] : \bar{C}$

By rule ($\Rightarrow$E) we have $\Gamma' \cup R(\Delta) \cup R'(\Delta' \cup R''(\Delta)) \vdash_k (v\,w)[x := u] : \bar{B}$. By the weakening lemma (cf. Lemma 16), it suffices to show that $\Gamma' \cup R(\Delta) \cup R'(\Delta') \cup R'(R''(\Delta)) \subseteq \Gamma \cup R(\Delta)$. This follows from $\Gamma = \Gamma' \cup R'(\Delta')$ and $R' \circ R'' \subseteq R$. ◀

▶ **Lemma 24** (Subject Reduction). *If* $\Gamma \vdash_k t : \bar{B}$ *and* $t \to_\beta u$ *then* $\Gamma \vdash_k u : \bar{B}$.

Proof. The proof is by induction over t.

Case $t = x$**:** This case is impossible since x is in β-normal form.

Case $t = \lambda x.v$**:** In this case, reduction actually has the form $t = \lambda x.v \to_\beta \lambda x.v' = u$ where $v \to_\beta v'$. The only way to derive type for $\lambda x.v$ is to use rule ($\Rightarrow$I). This means that $\bar{B} = \bar{\sigma} \Rightarrow \bar{A}$ for some $\bar{\sigma}$ and $\bar{A}$. From that we get a derivation for $\Gamma, x : \bar{\sigma} \vdash_k v : \bar{A}$. We can now use the induction hypothesis to get $\Gamma, x : \bar{\sigma} \vdash_k v' : \bar{A}$ and then rule ($\Rightarrow$I) to get derivability of $\Gamma \vdash_k v' : \bar{\sigma} \Rightarrow \bar{A}$, which is the required conclusion.

Case $t = v\,w$**:** In this case reduction has two forms. In the first of them, we reduce so that $v\,w \to_\beta v'\,w'$ where $v \to_\beta v'$ and $w = w'$ or $v = v'$ and $w \to_\beta w'$. To handle this situation, we can use decomposition by means of rule ($\Rightarrow$E) then the induction hypothesis and assemble the necessary derivation again by rule ($\Rightarrow$E).

In the second case $v = \lambda x.v'$ and reduction follows the β-redex $(\lambda x.v')\,w \to_\beta v'[x := w]$. Observe that the derivation of $\Gamma \vdash_k (\lambda x.v')\,w : \bar{B}$ must end with rule ($\Rightarrow$E)

$$\frac{\Gamma \vdash_k \lambda x.v' : R(\bar{A}) \Rightarrow \bar{B} \qquad \Delta \vdash_k w : \bar{A}}{\Gamma \cup R(\Delta) \vdash_k (\lambda x.v')\,w : \bar{B}}\ (\Rightarrow\text{E})$$

for some relation R and vector $\bar{A}$. Then rule ($\Rightarrow$I) must be used to derive the judgment $\Gamma \vdash_k \lambda x.v' : R(\bar{A}) \Rightarrow \bar{B}$, which gives derivation for $\Gamma, x : R(\bar{A}) \vdash_k v' : \bar{B}$. We can now use the substitution lemma (cf. Lemma 23) to obtain $\Gamma \cup R(\Delta) \vdash_k v'[x := w] : \bar{B}$, which is the desired conclusion. ◀

3 Relation to Traditional Intersection Type Systems

As a technical tool to establish a relation with the traditional intersection type systems, we use the so-called *strict* system of van Bakel [22, Section 5]. This clarifies the relationship with the system of Barendregt, Coppo, Dezani-Ciancaglini [2], which is expressed in Theorem 30. In addition, this helps to establish metatheoretical properties, such as normalization.

▶ **Definition 25** (Strict Intersection Type System [22, Definition 5.1])**.**
The rules of the strict system are slightly adapted to notation used in this paper.

$$\frac{A \in \sigma}{\Sigma, x : \sigma \vdash_S x : A}\ (\cap\text{E}) \qquad \frac{\Sigma \vdash_S t : A_1, \ldots, \Sigma \vdash_S t : A_n}{\Sigma \vdash_S t : \{A_1, \ldots, A_n\}}\ (\cap\text{I})$$

$$\frac{\Sigma, x : \sigma \vdash_S t : A}{\Sigma \vdash_S \lambda x.t : \sigma \to A}\ (\to\text{I}) \qquad \frac{\Sigma \vdash_S t : \sigma \to B \qquad \Sigma \vdash_S u : \sigma}{\Sigma \vdash_S t\,u : B}\ (\to\text{E})$$

▶ **Lemma 26.** *If* $\Gamma_{[i]} \vdash_S t : \bar{A}_{[i]}$ *for each* $i \in \{1, \ldots, \dim(\bar{A})\}$ *then* $\Gamma \vdash t : \bar{A}$.

Proof. Induction on t. Let $n := \dim(\bar{A})$ and assume $\Gamma_{[i]} \vdash_S t : \bar{A}_{[i]}$ for each $i \in \{1, \ldots, n\}$.

Case $t = x$**.** If $\Gamma_{[i]} \vdash_S x : \bar{A}_{[i]}$ for $i \in \{1, \ldots, n\}$ then each $\Gamma_{[i]}$ contains declaration $x : \bar{\sigma}_{[i]}$ such that $\bar{A}_{[i]} \in \bar{\sigma}_{[i]}$. Consequently, $\bar{A} \mathrel{\bar{\in}} \bar{\sigma}$ and we can use rule (Ax) to get $\Gamma \vdash t : \bar{A}$.

Case $t = \lambda x.u$**.** If $\Gamma_{[i]} \vdash_S \lambda x.u : \bar{A}_{[i]}$ for $i \in \{1, \ldots, n\}$, then these judgments must be the result of ($\to$I) rule. Consequently, $\bar{A} = \bar{\sigma} \Rightarrow \bar{B}$ for some $\bar{\sigma}$ and $\bar{B}$ and we have derivations for $\Gamma_{[i]}, x : \bar{\sigma}_{[i]} \vdash_S u : \bar{B}_{[i]}$. By the induction hypothesis, we obtain that $\Gamma, x : \bar{\sigma} \vdash u : \bar{B}$. We can now apply rule ($\Rightarrow$I) and obtain $\Gamma \vdash \lambda x.u : \bar{A}$.

Case $t = u\,v$**.** If $\Gamma_{[i]} \vdash_S u\,v : \bar{A}_{[i]}$ then this can only be derived by rule ($\to$E). Consequently, we have derivations for $\Gamma_{[i]} \vdash_S u : \bar{\sigma}_{[i]} \to \bar{A}_{[i]}$ and $\Gamma_{[i]} \vdash_S v : \bar{\sigma}_{[i]}$ for some $\bar{\sigma}$. Then the only way to derive $\Gamma_{[i]} \vdash_S v : \bar{\sigma}_{[i]}$ is to use rule ($\cap$I) so that $\Gamma_{[i]} \vdash_S v : B_i^j$ where for some

k_i we have $\bar{\sigma}_{[i]} = \{B_i^1, \dots, B_i^{k_i}\}$. Assume $\Gamma = \{x_1 : \bar{\sigma}^1, \dots, x_m : \bar{\sigma}^m\}$. By the induction hypothesis we construct a derivation for $\Delta \vdash v : \bar{B}$ where

$$\begin{aligned}
&\Delta := \{x_1 : \bar{\tau}^1, \dots, x_m : \bar{\tau}^m\} \\
&\bar{\tau}^i := (\underbrace{\bar{\sigma}^i_{[1]}, \dots \bar{\sigma}^i_{[1]}}_{k_1\text{times}}, \dots, \underbrace{\bar{\sigma}^i_{[n]}, \dots \bar{\sigma}^i_{[n]}}_{k_n\text{times}}) \quad \text{for } i \in \{1, \dots, m\} \\
&\bar{B} := (B_1^1, \dots, B_1^{k_1}, \dots, B_n^1, \dots, B_n^{k_n})
\end{aligned}$$

Let $l_0 := 0$ and $l_i := \sum_{j=1}^{i} k_j$ for $i \in \{1, \dots, n\}$. We define the relation $R := \{(p, q) \mid q \in \{1, \dots, n\} \text{ and } l_{q-1} < p \leq l_q\}$. Observe that

$$R(\bar{B})_{[j]} = \bar{\sigma}_{[j]} \quad \text{and} \quad R(\Delta(x_i)) = \bar{\sigma}^i \tag{1}$$

for $j \in \{1, \dots, n\}$ and $i \in \{1, \dots, m\}$. Note that $R(\Delta) = \Gamma$ and $R(\bar{B}) = \bar{\sigma}$ by observation (1) above. From the induction hypothesis we get $\Gamma \vdash u : \bar{\sigma} \Rightarrow \bar{A}$, and we can apply rule ($\Rightarrow$E) to obtain $\Gamma \vdash u\, v : \bar{A}$. ◀

▶ **Lemma 27.** *If $\Gamma \vdash t : \bar{A}$ then for each $i \in \{1, \dots, \dim(\bar{A})\}$ we have $\Gamma_{[i]} \vdash_S t : \bar{A}_{[i]}$.*

Proof. Induction on t. Let $n := \dim(\bar{A})$ and assume $\Gamma \vdash t : \bar{A}$.

Case $t = x$. If $\Gamma \vdash x : \bar{A}$ then it can be derived only by rule (Ax). Consequently, there is $(x : \bar{\sigma}) \in \Gamma$ such that $\bar{A} \mathrel{\bar{\in}} \bar{\sigma}$. Consequently, we can use rule ($\cap$E) to obtain $\Gamma_{[i]} \vdash_S x : \bar{A}_{[i]}$ for $i \in \{1, \dots, n\}$.

Case $t = \lambda x.u$. The judgment $\Gamma \vdash \lambda x.u : \bar{A}$ can be derived only by rule ($\Rightarrow$I) and thus $\bar{A} = \bar{\sigma} \Rightarrow \bar{B}$ for some $\bar{\sigma}$ and $\bar{B}$ and also $\Gamma, x : \bar{\sigma} \vdash u : \bar{B}$ is derivable. By the induction hypothesis, we obtain $\Gamma_{[i]}, x : \bar{\sigma}_{[i]} \vdash_S u : \bar{B}_{[i]}$ for each $i \in \{1, \dots, n\}$. Then rule ($\to$I) gives us derivability of $\Gamma_{[i]} \vdash_S \lambda x.u : \bar{\sigma}_{[i]} \to \bar{B}_{[i]}$. As $\bar{\sigma}_{[i]} \to \bar{B}_{[i]} = \bar{A}_{[i]}$, we obtain the claim.

Case $t = u\ v$. The judgment $\Gamma \vdash u\, v : \bar{A}$ can be derived only by rule ($\Rightarrow$E). Consequently, $\Gamma = \Gamma' \cup R(\Delta)$ for some Γ', Δ, and R such that $R(\bar{B})_{[i]} = \bar{\sigma}_{[i]}$ for $i \in \{1, \dots, n\}$, and $\Gamma' \vdash u : \bar{\sigma} \Rightarrow \bar{A}$ and $\Delta \vdash v : \bar{B}$ are derivable. By the induction hypothesis, we can derive

$$\begin{aligned}
&\Gamma'_{[i]} \vdash_S u : \bar{\sigma}_{[i]} \to \bar{A}_{[i]} \quad && \text{for } i \in \{1, \dots, n\} \\
&\Delta_{[j]} \vdash_S v : \bar{B}_{[j]} && \text{for } j \in \{1, \dots, \dim(\bar{B})\}
\end{aligned} \tag{2}$$

We have $\Gamma'(x) \mathrel{\bar{\cup}} R(\Delta)(x) \mathrel{\bar{\subseteq}} \Gamma(x)$ for $x \in \mathrm{dom}(\Gamma)$, and therefore $(\Delta(x))_{[j]} \subseteq (\Gamma(x))_{[i]}$ for $(j, i) \in R$. We can now apply the weakening lemma to derivations in (2) and obtain $\Gamma_{[i]} \vdash_S v : \bar{B}_{[j]}$ for $(j, i) \in R$, as well as $\Gamma_{[i]} \vdash_S u : \bar{\sigma}_{[i]} \to \bar{A}_{[i]}$ for $i \in \{1, \dots, n\}$. Using rule ($\cap$I) we derive $\Gamma_{[i]} \vdash_S v : \{\bar{B}_{[j]} \mid j\, R\, i\}$. Application of rule ($\to$E) gives $\Gamma_{[i]} \vdash_S u\, v : \bar{A}_{[i]}$ for $i \in \{1, \dots, n\}$. ◀

An important consequence of the lemma above is that many properties of van Bakel's system are inherited. In particular, system **R** has the strong normalization property for derivations without the empty vector, as van Bakel's system has [21, Theorem 7.5] for derivations without the empty type.

As a final step, we establish the connection with the system BCD of Barendregt, Coppo, Dezani-Ciancaglini [2]. The set of types in this system is bigger than the set of strict types of van Bakel. However, for each type φ in this system there are strict types $A_1, \dots, A_n$ such that φ is equivalent to $\{A_1, \dots, A_n\}$ (cf. [20, Lemma 2.2.2]). This equivalence is the canonical equivalence induced by the quasi-order ($\leq$) that is the subtyping relation in BCD system. Therefore, we do not lose any generality by writing $\Sigma \vdash_{\mathrm{BCD}} t : \sigma$, where Σ and σ comply to the syntax used in this paper. We can also write $\Sigma \vdash_{\mathrm{BCD}} t : A$ as strict types are a special case of BCD types. To establish the relation between system **R** and the system BCD, we use the following statement mentioned in [22, Section 5].

▶ **Theorem 28** (van Bakel, [22]). *If* $\Sigma \vdash_{\mathrm{BCD}} t : \sigma$ *then there are* $\Sigma', A_1, \ldots, A_n$ *such that* $\Sigma \leq \Sigma'$, $\Sigma' \vdash_S t : A_i$ *for each* $i \in \{1, \ldots, n\}$ *and* $\{A_1, \ldots, A_n\} \leq \sigma$.

Moreover, If $\Sigma \vdash_S t : A$ *then* $\Sigma \vdash_{\mathrm{BCD}} t : A$.

A similar theorem was proved [20, Theorem 2.2.6] for another variant of the strict system the rules of which are derivable in the strict system of van Bakel used in this paper.

Note that the formulation above omits the condition that t is in $\lambda\bot$-normal form (cf. [20]). This condition is trivially fulfilled as our terms do not use the constant $\bot$. In addition, the original formulation in [20] does not include the second part, which obtains a derivation in BCD from the one in the strict system. However, this part easily follows by observation that the rules of the strict system in its presentation in [22] are derivable in BCD system. We need one more result concerning the type ordering.

▶ **Proposition 29.** *If* $\sigma \leq \sigma'$ *then there is* $\tau \subseteq \sigma$ *such that* $|\tau| \leq |\sigma'|$ *and* $\tau \leq \sigma'$.

Proof. For the proof we use an observation made by Hindley, cf. [12, Lemma 3(i)], which also follows by easy induction from the work of Barendregt, Coppo, Dezani-Ciancaglini [2, Lemma 2.4(ii)]: *If* $\{A_1, \ldots, A_n\} \leq \{B_1, \ldots, B_m\}$ *then for each* $i \in \{1, \ldots, m\}$ *there is* $j \in \{1, \ldots, n\}$ *such that* $A_j \leq B_i$.

To prove the current proposition we observe that σ can be presented as $\{A_1, \ldots, A_n\}$ and σ' as $\{B_1, \ldots, B_m\}$. By the observation above, we take for each $i \in \{1, \ldots, m\}$ only one A_{j_i} such that $A_{j_i} \leq B_i$. We obtain then immediately that $\{A_{j_1}, \ldots, A_{j_m}\} \leq \sigma'$. ◀

The considerations above imply that derivability in system **R** corresponds to derivability in the system BCD up to top-level subtyping.

▶ **Theorem 30.**

1. *If* $\Gamma \vdash_k t : \bar{A}$ *then for* $i \in \{1, \ldots, \dim(\bar{A})\}$ *we have* $\Gamma_{[i]} \vdash_{\mathrm{BCD}} t : \bar{A}_{[i]}$.
2. *If* $\Sigma \vdash_{\mathrm{BCD}} t : \sigma$ *where* $\sigma = \{A_1, \ldots, A_n\}$ *then there are* $\Gamma, \bar{B}$ *with* $\dim(\Gamma) = \dim(\bar{B}) = n$ *such that* $\Sigma \leq \Gamma_{[i]}$, $\Gamma \vdash_k t : \bar{B}$ *for some* k *and* $\{\bar{B}_{[1]}, \ldots, \bar{B}_{[n]}\} \leq \sigma$.

Proof. For the proof of (1), we assume $\Gamma \vdash_k t : \bar{A}$, which by Lemma 27 gives us that $\Gamma_{[i]} \vdash_S t : \bar{A}_{[i]}$ for $i \in \{1, \ldots, \dim(\bar{A})\}$. Then the second statement of Theorem 28 gives us that $\Gamma_{[i]} \vdash_{\mathrm{BCD}} t : \bar{A}_{[i]}$.

For the proof of (2), we assume that $\Sigma \vdash_{\mathrm{BCD}} t : \sigma$. By the first statement of Theorem 28, we obtain an environment Σ' such that $\Sigma \leq \Sigma'$ and a type $\sigma' = \{B_1, \ldots, B_m\}$ such that $\Sigma' \vdash_S t : B_i$ for all $i \in \{1, \ldots, m\}$ and $\{B_1, \ldots, B_m\} \leq \sigma$ for some m (*). Note that we may assume that $m = n$. Indeed, when $n < m$ we just introduce more copies of some coordinate i, cf. Lemma 17. When $m < n$, we use Proposition 29 to obtain some $\tau \subseteq \sigma$ the size of which is not greater than the size of σ' and by the inclusion $\tau \subseteq \sigma$ we still have $\Sigma' \vdash_S t : B$ for all $B \in \tau$. This means that $\Sigma \vdash_S t : B_i$ for each $i \in \{1, \ldots, n\}$. Consider now $\bar{B} = (B_1, \ldots, B_n)$. We now define a context Γ so that $\mathrm{dom}(\Gamma) = \mathrm{dom}(\Sigma)$ and $(x : \bar{\rho}_x) \in \Gamma$ for $\bar{\rho}_x = (\underbrace{\rho, \ldots, \rho}_{n\text{-times}})$ where $(x : \rho) \in \Sigma'$. Note that $\Gamma_{[i]} = \Sigma'$ so Lemma 26 gives us that $\Gamma \vdash_k t : \bar{B}$ for some k. Note that $\{\bar{B}_{[1]}, \ldots, \bar{B}_{[n]}\} \leq \sigma$ by (*). ◀

4 Decidability of Inhabitation in Bounded Dimension

In this section, we give the main result of the present work: inhabitation in bounded dimension is decidable in 2-EXPTIME. This result is supplemented by the observation that for each fixed dimension inhabitation is decidable in EXPTIME.

▶ **Definition 31** (Inhabitation in Bounded Dimension). *Given* Γ, $\bar{A}$, *and* k *is the judgment* $\Gamma \vdash_k t : \bar{A}$ *derivable for some term* t*?*

The proof of decidability is constructed in three steps. First, we establish inversion properties (Lemma 32 and Lemma 33) for abstraction and iterated application. Second, we formulate a subformula property (Lemma 36) for β-normal form derivations to restrict the inhabitant search space. Third, we give an alternating exponential space inhabitant search algorithm, obtaining a 2-EXPTIME upper bound on inhabitation (Theorem 37).

▶ **Lemma 32.** *If* $\Gamma \vdash_k \lambda x.t : \bar{B}$, *then* $\bar{B} = \bar{\sigma} \Rightarrow \bar{A}$ *for some* $\bar{\sigma}$ *and* $\bar{A}$, *and* $\Gamma, x : \bar{\sigma} \vdash_k t : \bar{A}$.

Proof. Case analysis regarding the last rule application. ◀

▶ **Lemma 33.** *If* $\Gamma \vdash_k x\, u_1 \dots u_n : \bar{B}$ *then there exists* $\bar{A}_1, \dots, \bar{A}_n, R_1, \dots, R_n, \Delta_1, \dots, \Delta_n$ *such that*

1. $R_i \subseteq \{1, \dots, \dim(\bar{A}_i)\} \times \{1, \dots, \dim(\bar{B})\}$ *is left-total and* $\dim(\bar{A}_i) \leq k$ *for* $i \in \{1, \dots, n\}$
2. $R_1(\bar{A}_1) \Rightarrow \dots \Rightarrow R_n(\bar{A}_n) \Rightarrow \bar{B} \,\bar{\in}\, \Gamma(x)$
3. $R_i(\Delta_i) \subseteq \Gamma$ *for* $i \in \{1, \dots, n\}$
4. $\Delta_i \vdash_k u_i : \bar{A}_i$ *for* $i \in \{1, \dots, n\}$
5. $\Gamma \vdash_k x\, u_1 \dots u_i : R_{i+1}(\bar{A}_{i+1}) \Rightarrow \dots \Rightarrow R_n(\bar{A}_n) \Rightarrow \bar{B}$ *for* $i \in \{1, \dots, n\}$

Proof. Induction on n and case analysis regarding the last rule application. Left-totality of relations $R_1, \dots, R_n$ for Property (1) is established using Lemma 17. Property (5) is established using Lemma 16. ◀

With each type or a set of types we associate a set of subformulae (types or sets of types), by the following Definition 34.

▶ **Definition 34** (Subformula).

- $\mathrm{form}(a) := \{a\}$
- $\mathrm{form}(\sigma \to A) := \{\sigma \to A\} \cup \mathrm{form}(\sigma) \cup \mathrm{form}(A)$
- $\mathrm{form}(\sigma) := \{\sigma\} \cup \bigcup_{A \in \sigma} \mathrm{form}(A)$

The notion of occurring subformulae is extended to vectors:

- $\mathrm{form}(\bar{A}) := \bigcup_{i=1}^{\dim(\bar{A})} \mathrm{form}(\bar{A}_{[i]})$ *and* $\mathrm{form}(\bar{\sigma}) := \bigcup_{i=1}^{\dim(\bar{\sigma})} \mathrm{form}(\bar{\sigma}_{[i]})$

The number of subformulae of a given type or set of types is polynomial in the number of nodes in its syntax tree, illustrated by the following example.

▶ **Example 35.** We have $\mathrm{form}(\{a, b\} \to c) = \{\{a, b\} \to c, \{a, b\}, a, b, c\}$. In particular, the set $\{a, b\}$ and the types a, b are subformulae of $\{a, b\} \to c$, but the sets $\{a\}$ and $\{b\}$ are not.

▶ **Lemma 36** (Subformula Property). *Assume that* $\Gamma \vdash_k t : \bar{A}$ *such that* t *is in* β*-normal form. Let* $\mathcal{F} := \mathrm{form}(\bar{A}) \cup \bigcup\{\mathrm{form}(\bar{\tau}) \mid (y : \bar{\tau}) \in \Gamma\}$ *be the set of all subformulae of types and sets of types occurring in* Γ *and* $\bar{A}$. *There is a derivation of* $\Gamma \vdash_k t : \bar{A}$ *such that for each judgment* $\Delta \vdash_k u : \bar{B}$ *in this derivation we have*

1. *for* $(x : \bar{\sigma}) \in \Delta$ *and* $i \in \{1, \dots, \dim(\Delta)\}$ *we have* $\bar{\sigma}_{[i]} \in \mathcal{F} \cup \{\emptyset\}$
2. $\bar{B}_{[i]} \in \mathcal{F}$ *for* $i \in \{1, \dots, \dim(\bar{B})\}$

Proof. Induction on the size of t. There are two cases.

Case $t = \lambda x.u$: Follows immediately from Lemma 32 and the induction hypothesis.

Case $t = x\,u_1 \dots u_n$: There exist $\bar{A}_1, \dots, \bar{A}_n$, $R_1, \dots, R_n$, $\Delta_1, \dots, \Delta_n$ satisfying properties of Lemma 33(1) – Lemma 33(5). Judgments $\Gamma \vdash_k x\,u_1 \dots u_i : R_{i+1}(\bar{A}_{i+1}) \Rightarrow \dots \Rightarrow R_n(\bar{A}_n) \Rightarrow \bar{B}$ (cf. Lemma 33(5)) satisfy properties (1) and (2). While subderivations $\Delta_i \vdash_k u_i : \bar{A}_i$ for $i \in \{1, \dots, n\}$ satisfy Property (2), they do not necessarily satisfy Property (1). By Lemma 33(3) type assumptions in Δ_i for $i \in \{1, \dots, n\}$ may be strict subsets of subformulae in $\mathcal{F}$. By Lemma 16 and Lemma 8(7) we can weaken the type assumptions by saturating the sets of types to subformulae in $\mathcal{F}$. This results in derivation environments $\Delta'_1, \dots, \Delta'_n$ and subderivations of judgments $\Delta'_i \vdash_k u_i : \bar{A}_i$ for $i \in \{1, \dots, n\}$ satisfying properties (1) and (2). We obtain the claim by the induction hypothesis and repeated application of rule (⇒E). ◀

▶ **Theorem 37.** *Given Γ, $\bar{A}$, and k it is decidable in 2-EXPTIME whether the judgment $\Gamma \vdash_k t : \bar{A}$ is derivable for some term t.*

Proof. In order to decide whether $\Gamma \vdash_k t : \bar{A}$ is derivable for some t, by subject reduction (Lemma 24) and normalization properties [21, Theorem 6.5 and Theorem 7.5] (combined with Lemma 27), it suffices to consider t in β-normal form. In the following, we develop an AEXPSPACE decision procedure.

Let $\mathcal{F} := \mathrm{form}(\bar{A}) \cup \bigcup\{\mathrm{form}(\bar{\tau}) \mid (y : \bar{\tau}) \in \Gamma\}$ be the set of all subformulae of types and sets of types occurring in Γ and $\bar{A}$, and let $N := |\mathcal{F}|$ denote its cardinality. By Lemma 36, we can consider a derivation of $\Gamma \vdash_k t : \bar{A}$ such that for each judgment $\Delta \vdash_k u : \bar{B}$ in this derivation each coordinate of occurring type vectors can take at most $N + 1$ distinct values (including the empty set for sets of types in environments). Therefore, the number of distinct k-dimensional vectors to consider in derivations is bounded by $(N + 1)^k$.

The decision procedure works as follows. First, we non-deterministically guess whether t is of shape $\lambda x.u$ for some x and u, or of shape $x\,u_1 \dots u_n$ for some $x \in \mathrm{dom}(\Gamma)$ and some $u_1, \dots, u_n$ where n is bounded by the maximal nesting of the arrow type constructor to the right in types occurring in $\Gamma(x)$.

If t is of shape $\lambda x.u$, then by Lemma 32 we have $\bar{A} = \bar{\sigma} \Rightarrow \bar{A}'$ for some $\bar{\sigma}$ and $\bar{A}'$, and we recursively decide whether $\Gamma, x : \bar{\sigma} \vdash_k u : \bar{A}'$ holds for some u.

If t is of shape $x\,u_1 \dots u_n$, then we use Lemma 33, universally branching in $i \in \{1, \dots, n\}$. In each branch i, we non-deterministically guess the necessary relation R_i (bounded in cardinality by k^2), type vector $\bar{A}_i$, environment Δ_i, and recursively decide whether the judgment $\Delta_i \vdash_k u_i : \bar{A}_i$ is derivable for some u_i.

By the pigeonhole principle, if a type environment contains more than $(N + 1)^k$ variables, some must have identical assigned vectors. In the non-deterministic guess of the shape $\lambda x.u$, if x is assigned the vector $\Gamma(y)$ for some $y \in \mathrm{dom}(\Gamma)$, we can equivalently guess $\lambda x.u[x := y]$ instead and remove the assumption on x.

Overall, the space required by the alternating decision procedure is polynomial in $(N+1)^k$, yielding an AEXPSPACE, or equivalently, a 2-EXPTIME decision procedure. ◀

If we fix a dimension bound k, we obtain an EXPTIME inhabitation decision procedure.

▶ **Theorem 38.** *For any $k \in \mathbb{N}$, given Γ and $\bar{A}$ it is decidable in EXPTIME whether the judgment $\Gamma \vdash_k t : \bar{A}$ is derivable for some term t.*

Proof. Similarly to the proof of Theorem 37, the number of distinct vectors in recursive invocation of the decision procedure is $(N + 1)^k$, which is polynomial for any fixed k. This yields an APSPACE, or equivalently, an EXPTIME decision procedure. ◀

5 Comparison with the Multiset-dimensional System

Comparing the dimensionally bounded system $\mathbf{R}$ to the currently most expressive dimensionally bounded system $(\Vdash_k)$ with decidable inhabitation [7, Section 3.2], one can observe several differences.

The system $(\Vdash_k)$ introduces non-idempotent intersection types (multisets to the left of the arrow) as annotations throughout the derivation in order to capture intersection introduction *as a resource*. This is different from traditional non-linear systems [4] and induces additional bookkeeping (see side condition $(\star)$ [7, Section 3.2] in rule $(\cap\mathrm{I})$). The exact correspondence to established intersection type systems (both idempotent and non-idempotent) is not immediate.

The most important difference is that the individual dimensions in system $(\Vdash_k)$ are strictly less comprehensive with respect to inhabited terms. First, any derivation in $(\Vdash_k)$ can be translated to a derivation in $(\vdash_k)$.

▶ **Lemma 39.** *If* $\Gamma_{[1]} \Vdash_k t : A$ *and* $\dim(\Gamma) = 1$*, then* $\Gamma \vdash_k t : (A)$.

Proof. Both $(\Vdash_k)$ and $(\vdash_k)$ follow derivations in $(\vdash_S)$. Using lexicographic ordering, multisets of cardinality at most k at top-level can be captured by vectors of size at most k. Therefore, the same construction as the proof of Lemma 26 applies to the translation. ◀

Since inhabitation in $(\Vdash_k)$ is EXPSPACE-complete [7, Theorem 34], the above Lemma 39 implies EXPSPACE-hardness of inhabitation in $(\vdash_k)$.

▶ **Corollary 40.** *Given* Γ*,* $\bar{A}$*, and* k *it is EXPSPACE-hard to decide whether the judgment* $\Gamma \vdash_k t : \bar{A}$ *is derivable for some term* t*.*

Second, the following Lemma 41 shows that there are types $A_1, A_2, \ldots$ which are inhabited in dimension 2, but each A_i requires multiset dimension 2^i to be assigned to any term.

▶ **Lemma 41.** *There exists a family of types* $\{A_1, A_2, \ldots\}$ *such that for all* $n \in \mathbb{N}$*:*

1. *The size (number of nodes in the syntax tree) of* A_n *is linear in* n*.*
2. *There exists a term* t_n *such that* $\emptyset \vdash_2 t_n : (A_n)$*.*
3. *If there exists a term* u_n *such that* $\emptyset \Vdash_k u_n : A_n$*, then* $k \geq 2^n$*.*

Proof. Let $\sigma_n := \bigcup_{i=1}^{n} \{\{a_i, b_i\} \to a_{i+1}, \{a_i, b_i\} \to b_{i+1}\}$, $\tau_n := \{\{a_{n+1}, b_{n+1}\} \to c\}$ and $A_n := \{a_0, b_0\} \to \sigma_n \to \tau_n \to c$.

1. This property holds by definition.
2. Let $v_0 := x$ and $v_{i+1} := y\, v_i$ for $i \in \{0, \ldots, n\}$ and let $t_n := \lambda xyz.z\, v_n$. We have $\emptyset \vdash_2 t_n : (A_n)$. The crucial iterated step for $i = 1, \ldots n$ is

$$\frac{\Gamma \vdash_2 y : R((a_i, b_i)) \Rightarrow (a_{i+1}, b_{i+1}) \qquad \Gamma \vdash_2 v_i : (a_i, b_i)}{\Gamma \vdash_2 y\, v_i : (a_{i+1}, b_{i+1})}\ (\Rightarrow\mathrm{E})$$

   Where for the environment $\Gamma := \{x : (\{a_0, b_0\}, \{a_0, b_0\}), y : (\sigma_n, \sigma_n), z : (\tau_n, \tau_n)\}$ and the relation $R := \{(1,1), (1,2), (2,1), (2,2)\}$ we observe that $R(\Gamma) = \Gamma$.
3. An inhabitant of A_n is necessarily of shape $\lambda xyz.z\,(y \ldots (y\, x) \ldots)$ such that y occurs at least n times and each occurrence of y doubles the annotation multiset cardinality. ◀

Third, complementarily to the above lemma, there is a family of terms which are typed at a specific type A in dimension 2, but require an exponentially large multiset dimension.

▶ **Lemma 42.** *There exists a type A and a family of terms $\{t_1, t_2, \ldots\}$ such that for all $n \in \mathbb{N}$:*

1. *The size (number of nodes in the syntax tree) of t_n is linear in n.*
2. *$\emptyset \vdash_2 t_n : (A)$.*
3. *If $\emptyset \Vdash_k t_n : A$, then $k \geq 2^n$.*

Proof. Let $u_0 := x$ and $u_{n+1} := y\, u_n$ for $n \in \mathbb{N}$, and let $t_n := \lambda xyz.z\, u_n$. Additionally, let $\sigma := \{\{a,b\} \to a, \{a,b\} \to b\}$, let $\tau := \{\{a,b\} \to c\}$, and let $A := \{a,b\} \to \sigma \to \tau \to c$.

1. This property holds by definition.
2. We have $\emptyset \vdash_2 t_n : (A)$. The crucial step for $i \in \mathbb{N}$ is

$$\frac{\Gamma \vdash_2 y : R((a,b)) \Rightarrow (a,b) \qquad \Gamma \vdash_2 u_i : (a,b)}{\Gamma \vdash_2 y\, u_i : (a,b)}\ (\Rightarrow\mathrm{E})$$

   Where $\Gamma := \{x : (\{a,b\}, \{a,b\}), y : (\sigma, \sigma), z : (\tau, \tau)\}$ and $R := \{(1,1), (1,2), (2,1), (2,2)\}$, observing that $R(\Gamma) = \Gamma$.
3. Similarly to the proof of Lemma 41, each occurrence of y via rule ($\cap$I) doubles the annotation multiset cardinality. ◀

The avid reader may wonder: why is the EXPSPACE inhabitation procedure $\mathcal{A}_{\langle d \rangle}$ [7], which respects a bound d on the multiset dimension, not suited for inhabitation in dimensionally bounded system **R**? One immediate answer is given by the construction in the proof of Lemma 41. Iterative intersection introduction strictly increases multiset dimension. This is reflected in procedure $\mathcal{A}_{\langle d \rangle}$ via *multisets of simultaneous constraints* with strictly increasing cardinalities on intersection introduction.

Still, the immediate answer is not satisfactory. The construction in the proof of Lemma 41 can easily be accommodated for by identifying redundant constraints. The modified inhabitation procedure would remain in EXPSPACE.

A key property of procedure $\mathcal{A}_{\langle d \rangle}$ is that environments in simultaneous constraints are monotone with respect to pointwise inclusion. That is, there is no need to strengthen type assumptions. This is different for the system **R** for which in rule ($\Rightarrow$E) type assumptions of premises are combined, including the action of a relation R. This violates the monotonicity condition, and the EXPSPACE (or, alternating exponential time) argument is not applicable.

Overall, the present notion of dimension lies strictly between the multiset dimension (EXPSPACE-complete inhabitation [7, Theorem 34]) and set dimension (undecidable inhabitation [7, Theorem 28]), while preserving decidability of inhabitation.

6 Conclusion

We introduced a new syntax-directed intersection type system, which we call system **R**, related to traditional systems [2, 21, 22], where typing judgments assign vectors of types rather than single types. Special relations control information flow between vector components, allowing precise tracking of information flow while keeping derivations syntax-directed and manageable. Vectors also enable a natural restriction on derivations: only vectors of dimension at most k are allowed, which is crucial for later results.

This system fulfills the need for a formalism in which natural bound on a resource, here, the number of information-flow channels corresponding to vector dimension, renders inhabitation decidable. With the maximal dimension being part of the input, inhabitation is 2-EXPTIME decidable, and fixing it as a parameter yields an EXPTIME decision procedure. Without a bound, inhabitation becomes undecidable [19], as typings in our system coincide with ones in traditional systems. We also establish EXPSPACE-hardness when the maximal dimension is part of the input, leaving the exact complexity as an open problem.

An earlier system [7] imposes a similar dimensional restriction. We therefore provided a detailed comparison highlighting both conceptual similarities and key differences in expressiveness and technical treatment, throughout which demonstrating that system **R** offers a simpler and strictly more comprehensive framework that is easier to use in practice.

Further work. Determining the precise complexity of inhabitation in the setting of system **R** remains an open and challenging problem. Addressing it would require resolving subtle combinatorial issues that arise from the interplay of bounded resources and vector-based type structures, making this a promising direction for future research.

Another important avenue of investigation concerns type inference for system **R**. Since the system incorporates features of both set and multiset interpretations of types, it is not immediately clear whether the typability problem falls into NP or PSPACE, as established for related systems by Dudenhefner and Rehof [8]. Studying this question could provide deeper insight into the algorithmic behavior of the system. Similarly to system **Rel** of de Carvalho [6, Definition 4.1], it is interesting whether binary relations in system **R** can be related to term evaluation by means of a Krivine machine. As the system λ^{CIL} by Wells et al. [23] was also developed in the context of type inference problems, it would be interesting to compare it with the current system on the technical level.

As the system **Rel** of de Carvalho uses relations in its models, it is interesting to see how the binary relations in the current system **R** fit into this kind of denotational semantics.

Finally, intersection type systems are often extended with subtyping, so developing a version of system **R** with subtyping presents another interesting direction. Such an extension could potentially support subject expansion and reduction under η-reduction, but it would also complicate the algorithmic analysis of type inference and inhabitation problems, requiring careful treatment to balance expressiveness and decidability.

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